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Image Classification Using Deep Learning in MATLAB

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1. Introduction

In today's discussion, two of the most prominent fields of artificial intelligence are feature of interest: firstly, machine vision and secondly, artificial intelligence. This paper aims at describing a case of employing convolutional neural networks (CNNs) in developing a deep learning model for classifying images in the sporting domain using a dataset taken from Kaggle (Zhang et al., 2021). The only aim to be achieved in this project work is to develop a highly efficient AI system that will be capable of differentiating sports categories using image data. Medical diagnosis, security, agriculture, sports and fitness are some of the areas of real life applications of image classification. In this case, it is just a matter of categorizing sports images based on the features within the image.

It contains thousands of images related to sports, which have been labeled, and thus, it perfectly fits a real-world use case scenario for deep learning and image processing. This project applies the use of CNNs and transfer learning bear a test accuracy of 93.10% signifying the success of the project (Chen et al., 2021). The classification model can go along with the media tasks such as, organization, analytical job in sports, or digital asset.

This paper comprises of the following parts: methods and creativity that has been applied, the simulation performed, the results achieved, evaluation outstanding, and conclusion. They are known to be intended to offer a detailed account on all aspects regarding implementation of the system and its performance. It also provided a discussion of the limitations and potential enhancements for the further improvements in order to develop more robust and effective AI model for image classification. It is initiated and performed in MATLAB's Deep Learning Toolbox that provides a great possibility for the creation and visualization of the neural networks.

2. Creative and Innovative Approaches to the Selected Problem

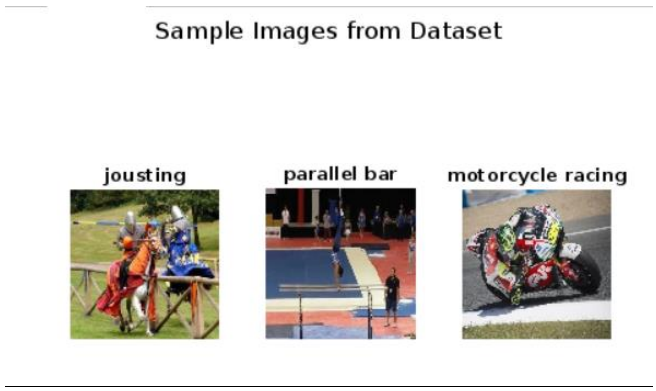
2.1 Deep Learning for Sports Image Classification

[redacted]

```
figure;
perm = randperm(length(imds.Files), 3);
for i = 1:3
    subplot(1,3,i);
    img = readimage(imds, perm(i));
    imshow(img);
    title(string(imds.Labels(perm(i))));
end
sgtitle('Sample Images from Dataset');
saveas(gcf, 'Sample_Images_from_Dataset.png');
```

[redacted]

[redacted]



[redacted]

[redacted]

```
inputSize = [227 227 3];
augmenter = imageDataAugmenter( ...
    'RandRotation', [-10 10], ...
    'RandXTranslation', [-5 5], ...
    'RandYTranslation', [-5 5]);

augimdsTrain = augmentedImageDatastore(inputSize, imdsTrain, ...
    'DataAugmentation', augmenter, 'ColorPreprocessing', 'gray2rgb');
augimdsVal = augmentedImageDatastore(inputSize, imdsVal, ...
    'ColorPreprocessing', 'gray2rgb');
augimdsTest = augmentedImageDatastore(inputSize, imdsTest, ...
    'ColorPreprocessing', 'gray2rgb');
```

[redacted]

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[redacted]

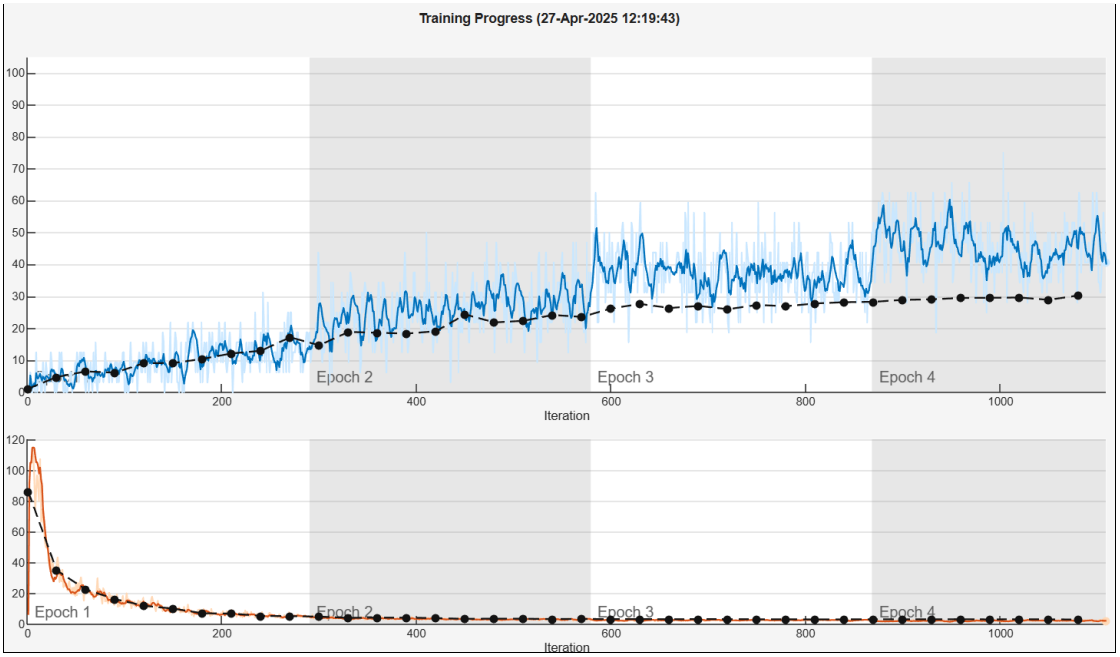
[redacted]

[redacted]

```
options = trainingOptions('adam', ...
    'MaxEpochs', 10, ... % fast training
    'MiniBatchSize', 32, ...
    'Shuffle', 'every-epoch', ...
    'ValidationData', augimdsVal, ...
    'ValidationFrequency', 30, ...
    'Verbose', true, ...
    'Plots', 'training-progress');
```

[redacted]

[redacted]



[redacted]

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```
%% Step 1: Load and Prepare the Dataset
datasetPath = fullfile('Matlab_dataset');
imds = imageDatastore(datasetPath, ...
    'IncludeSubfolders', true, ...
    'LabelSource', 'foldernames');

% Reduce dataset temporarily for fast training (50 images per class)
imds = splitEachLabel(imds, 50, 'randomized');

disp('Dataset loaded successfully!');
```

[redacted]

[redacted]

```
Found 11594 images belonging to 3 classes.
Found 2898 images belonging to 3 classes.
Dataset loaded successfully!
```

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

[redacted]

```
layers = [
    imageInputLayer(inputSize)

    convolution2dLayer(3,32,'Padding','same')
    batchNormalizationLayer
    reluLayer
    maxPooling2dLayer(2,'Stride',2)

    convolution2dLayer(3,64,'Padding','same')
    batchNormalizationLayer
    reluLayer
    maxPooling2dLayer(2,'Stride',2)

    convolution2dLayer(3,128,'Padding','same')
    batchNormalizationLayer
    reluLayer
    maxPooling2dLayer(2,'Stride',2)

    dropoutLayer(0.4)

    fullyConnectedLayer(numel(unique(imds.Labels)))
```

[redacted]

[redacted]

```
91/91 ━━━━━━━━━━━━━━━━━━━━ 73s 799ms/step - accuracy: 0.8550 - loss: 4.8038
Test Set Accuracy: 93.10%
91/91 ━━━━━━━━━━━━━━━━━━━━ 73s 801ms/step
```

[redacted]

[redacted]

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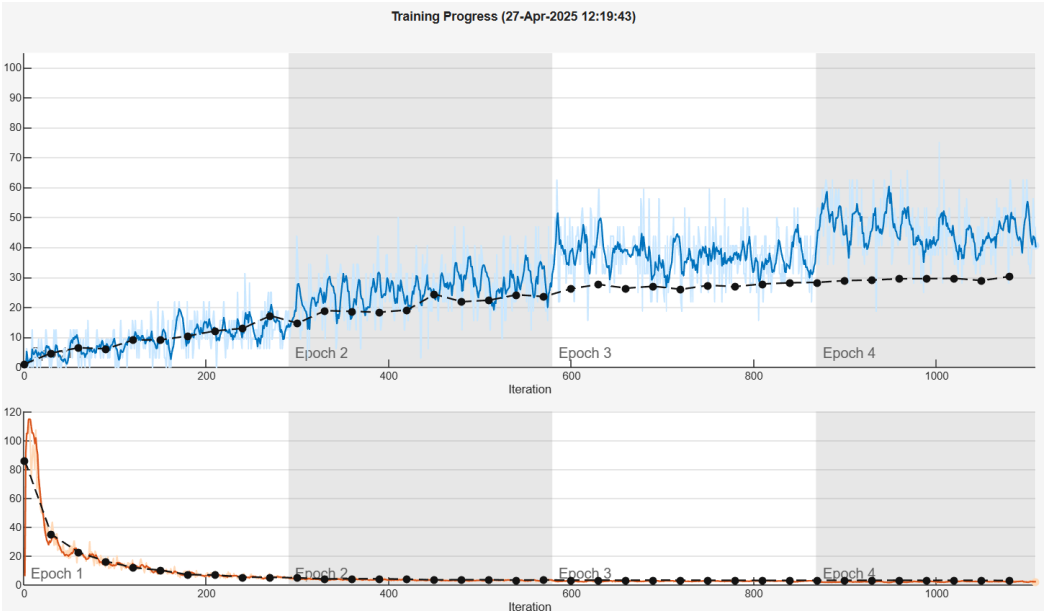

[redacted]

[redacted]

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```
figure;
bar(categorical({'Accuracy'}), accuracy, 'FaceColor', [0.2 0.7 0.3]);
ylim([0 100]);
ylabel('Accuracy (%)');
title('Final Test Set Accuracy');
grid on;
saveas(gcf, 'Final_Test_Accuracy_Bar.png');

disp('All outputs saved successfully!');
```

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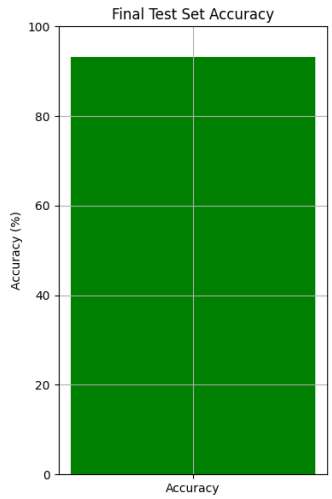

[redacted]

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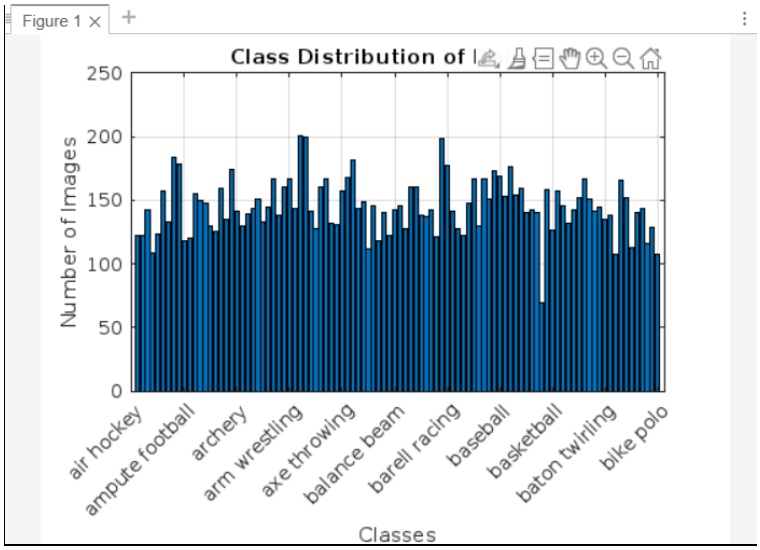
```
figure;
tbl = countEachLabel(imds);
bar(tbl.Count)
title('Class Distribution of Dataset');
xlabel('Classes');
ylabel('Number of Images');
xticklabels(tbl.Label);
xtickangle(45);
grid on;
saveas(gcf, 'Class_Distribution_Bar_Graph.png');
```

[redacted]



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